

Experiment No. 3

Aim: To determine the molarity of the given solution of sodium hydroxide (NaOH) by titrating it against standard solution of (0.05M) oxalic acid. ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$).

Apparatus: Burette, pipette, conical flask, burette stand, white porcelain tile, etc.

Chemicals: Standard solution of (0.05 M) oxalic acid, NaOH solution, phenolphthalein indicator.

Principle: This is an alkalimetry type of titration. The titration is in between weak acid ($\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$) and strong base (NaOH).

Procedure:

1. Wash the burette, pipette and conical flask with water.
2. Rinse the burette with given NaOH solution and then fill the burette with NaOH solution. Remove the air bubble from the nozzle and avoid leakage if any.
3. Adjust the level of the solution up to zero mark and fix it on a burette stand.
4. Rinse the pipette with standard oxalic acid solution.
5. Pipette out 10 mL of 0.05M oxalic acid solution and put it carefully in a clean, empty conical flask. Then add 2 drops of phenolphthalein indicator. (The solution remains colorless.)
6. Place the conical flask on a white porcelain tile below the burette.
7. Start adding NaOH solution dropwise from the burette in conical flask with continuous stirring, till light / faint pink color is obtained in conical flask. Stop adding NaOH, this is the end point of titration.
8. Keep the flask constantly shaking in a circular manner, this is called swirling of solution.
9. See that the color does not disappear even on vigorous shaking. This is the end point titration.
10. Note this reading as 'Pilot' reading. (Pilot reading is always in whole numbers.)
11. Now fill the burette again with NaOH solution up to the zero mark.
12. Repeat the above procedure and take minimum three more correct burette readings.
13. Note down the constant burette reading 9.8 'x' mL (C.B.R.)
14. From C.B.R. calculate the molarity of NaOH solution.

Observation:

1. Solution in a burette : Unknown NaOH Solution
2. Solution by a pipette : 10 ml (0.05 M) oxalic acid
3. Indicator : phenolphthalein indicator
4. End point : colourless to light pink
5. Chemical Equation : $\text{H}_2\text{C}_2\text{O}_4 + 2\text{NaOH} \longrightarrow \text{Na}_2\text{C}_2\text{O}_4 + 2\text{H}_2\text{O}$

Observation Table:

Burette reading	Pilot reading	Burette reading in mL			C.B.R.
		I	II	III	
Final	9	9.8	9.8	9.8	9.8 (x) mL
Initial	to 10 mL	0.0	0.0	0.0	
Difference		9.8	9.8	9.8	

Calculations: From above chemical equation

2 mol of NaOH combines with 1 mol of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$

$$\therefore 80 \text{ g NaOH} = 126 \text{ g } \text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$$

(Molar mass of NaOH = 40 g/mol, Molar mass of $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ = 126 g/mol)

1. To calculate 'g' of NaOH

$$\therefore 1000 \text{ mL } 1\text{M } \text{H}_2\text{C}_2\text{O}_4 \equiv 80 \text{ g NaOH}$$

$$\therefore 10 \text{ mL } 1\text{M } \text{H}_2\text{C}_2\text{O}_4 \equiv 8 \text{ g NaOH}$$

$$\therefore 10 \text{ mL } 0.05\text{M } \text{H}_2\text{C}_2\text{O}_4 \equiv 0.4 \text{ g NaOH}$$

Therefore, 0.4 g NaOH is present in (x) mL of NaOH solution of unknown molarity.

$$\therefore \text{..... (x) mL NaOH solution contains 0.4 g NaOH}$$

$$\therefore 1000 \text{ mL NaOH solution contains} = \frac{1000 \times 0.4}{\text{..... (x)}} \text{ g of NaOH}$$

2. To calculate molarity of NaOH

Space for calculation

$$\therefore \text{Molarity of NaOH} = \frac{1000 \times 0.4}{\text{..... (x)} \times 40}$$

$$= \frac{1}{\text{..... (x)}}$$

$$= 0.1020 \text{ M [By reciprocal]}$$

Result:

Solution	Molarity
NaOH	0.1020 M

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type of questions.

1. A chemical substance indicates completion of the reacting solution by sharp change in its color is called
a. catalyst
c. an inhibitor
b. ☒ an indicator
d. reagent
2. Sodium oxalate is salt of
a. strong acid and weak base
c. ☒ weak acid and strong base
b. weak acid and weak base
d. strong acid and strong base
3. In the volumetric analysis of NaOH against 0.05M $\text{H}_2\text{C}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ constant burette reading was found to be 9.8 mL. The molarity of NaOH is
a. ~~0.1020 M~~
c. 0.01200 M
b. 0.1120 M
d. 0.0980 M
4. Indicator used in the titration of weak acid and strong base is
a. methyl orange
c. litmus
b. ☒ phenolphthalein
d. universal indicator
5. When certain volume of one solution completely reacts with the other solution, this stage of titration is called
a. equivalence point
c. ☒ end point
b. constant burette reading
d. initial burette reading

Short answer questions

1. Why molarity is a temperature dependent quantity?

Molarity is a temperature dependent quantity because it is defined as the number of moles of solute per litre of solution. Since the volume of a solution changes with temperature, the molarity also changes.

2. During acid base titration by using phenolphthalein as an indicator, why the color of the reaction mixture changes from colorless to faint pink?

Phenolphthalein is colourless in acidic medium and pink in basic medium. At the end point of the titration, a slight excess of base makes the solution slightly alkaline, causing the colour to change from colourless to faint pink.

3. Calculate the molar mass of the following compounds?

a. H_2SO_4 b. Na_2CO_3 (Atomic mass: C=12, H=1, O=16, S=32 and Na=23)

Ans

a) HSO_4^-

Molar mass = $2(1) + 32 + 4(16)$
= $2 + 32 + 64$
= 98 g mol^{-1}

b) Na_2CO_3

Molar mass = $2(23) + 12 + 3(16)$
= $46 + 12 + 48$
= 106 g mol^{-1}

4. Define a) molarity b) titrant

Ans a.

Molarity is the number of moles of solute dissolved in one litre of solution.

b.

The standard solution taken in the burette and added during titration is called the titrant.

Remark and sign of teacher: _____

Molarity = $\frac{\text{no. of moles of solute}}{\text{volume of solution in dm}^3}$